Complex Analysis

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Abstract

This paper covers review of the Complex Analysis, with the concept of Homotopy, Bounded Variation,

Contents

1 Basic Definitions

Definition 1.1. Given distinct two points $a, b \in \mathbb{C}$, define *the line segment*

$$[a, b] \stackrel{\text{def}}{=} \{(1-t)a + tb \mid 0 \leq t \leq 1\}$$

And define the *multiple line segment* as

$$[a, b, c] \stackrel{\text{def}}{=} [a, b] \cup [b, c]$$

Define the *length* of the line segement:

$$\text{length}[a, b] \stackrel{\text{def}}{=} |b - a|$$

and

$$\text{length}[a, b, c] \stackrel{\text{def}}{=} \text{length}[a, b] + \text{length}[b, c]$$

The definitions followed by Stein, Complex Analysis.

Definition 1.2. Let G be an open set in $\mathbb{C}$. A map $f : G \to \mathbb{C}$ is called *holomorphic* at $a \in G$ if: the limit

$$f'(a) \stackrel{\text{def}}{=} \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists.

2 Line Integral

2.1 Definition

Definition 2.1. Let $\gamma : [a,b] \to \mathbb{C}$ be a bounded variation(or rectifiable), and $f : \mathbb{C} \to \mathbb{C}$ be a continuous on $\gamma[a,b]$. Define the *Line Integral* of f along γ is:

$$\int_\gamma f(z)dz \overset{\text{def}}{=} \int_a^b f d\gamma$$

Lemma 2.2. If $\gamma : [a,b] \to \mathbb{C}$ is piecewise smooth, then γ is of bounded varation with

$$TV(\gamma) = \int_a^b |\gamma'(t)|dt$$

Theorem 2.3. Let $\gamma : [a,b] \to \mathbb{C}$ be of bounded variation and $f : [a,b] \to \mathbb{C}$ be a continuous map.

3 Goursat's Theorem

Theorem 3.1. Let G be an open set and let $f : G \to \mathbb{C}$ be a holomorphic function. Then,

$$\int_T f = 0$$

for every triangular path T in G.

Proof. Let $T_0 = [a, b, c, a]$ be the triangular path in G, and let $d = \operatorname{diam} T$ and $L = \operatorname{length} T$.
And, construct four subtriangles

$$T^{(1)} = \left[a, \frac{a+b}{2}, \frac{a+c}{2}, a\right], \; T^{(2)} = \left[\frac{a+b}{2}, b, \frac{b+c}{2}, \frac{a+b}{2}\right], \; T^{(3)} = \left[\frac{a+c}{2}, \frac{b+c}{2}, c, \frac{a+c}{2}\right], \; T^{(4)} = \left[\frac{a+b}{2}, \frac{b+c}{2}, \frac{a+c}{2}, \frac{a+b}{2}\right]$$

Then,

$$\int_{T_0} f = \int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f$$

Thus, by the triangle inequailty,

$$\left|\int_{T_0} f\right| \le \left|\int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f\right| \le 4 \cdot \max_{i=1,2,3,4} \left|\int_{T_i} f\right|$$

Let $T_1 = T^j$, where the j maximizes the above integral. Apply the same process for $T_0, T_1, \ldots, T_{n-1}$, we obtain that a sequence of nested compact sets $\{\mathcal{T}_n\}$, where $\mathcal{T}_n$ is the convex hull of T_n. So, the intersection

$$\bigcap_{n=0}^{\infty} \mathcal{T}_n = \{z_0\}$$

is a singleton. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies \left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon$$

since f is holomorphic. Moreover, by the construction, there exists $N \in \mathbb{N}$ such that

$$T_N \subseteq B(z_0; \delta)$$

Now, the triangle inequality again,

$$\left|\int_{T_N} f(z)dz\right| = \left|\int_{T_N} f(z) - f(z_0) - f'(z_0)(z - z_0)dz\right| \le \varepsilon \int_{T_N} |z - z_0||dz| = \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N$$

Finally, since $\operatorname{diam} T_N = 2^{-n} \operatorname{diam} T_0$ and $\operatorname{length} T_N = 2^{-n} \operatorname{length} T_0$,

$$\left|\int_{T_0} f(z)dz\right| \le 4^N \cdot \left|\int_{T_N} f(z)dz\right| \le 4^N \cdot \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N \le \varepsilon \cdot \operatorname{diam} T_0 \cdot \operatorname{length} T_0 \to 0 \;\text{ as }\; \varepsilon \to 0$$

$\square$

4 Morera's Theorem

Theorem 4.1. Let G be a region and let $f : G \to \mathbb{C}$ be a continuous function such that

$$\int_T f = 0$$

for every triangular path T in G. Then, f is analytic in G.

Cauchy-Goursat’s Theroem